

ADL v0.92 — Code Review

Production birthday activation path · operator-requested, outside the WP-23 packet

Date	2026-08-24
Requested by	Operator, in chat — not a WP-23/#312 deliverable
Packet scope note	THIRD_PARTY_REVIEW_HANDOFF_v0.92.md excludes code review (“New product implementation or behavioral repair”). File separately from the documentation findings.
Files read	adl/src/production_birthday.rs; adl-runtime-kernel/src/production_birthday.rs; adl-runtime-kernel/src listing
Assessment	Strongest code read in this project to date. 3 low-severity findings, 0 P0/P1.
Executed	Nothing. Read, not run.

Method and honesty boundary

Two files were read line-level and one crate listing was measured. No build, no test execution, no failpoint injection, no cargo audit, no git operations. The failpoint tests referenced below exist as an API surface in the source; whether they pass at this revision was not verified here. Nothing in this document asserts runtime behaviour — only what the source says. This is a narrow review of one path, not a crate-wide or workspace-wide assessment, and confers no release credit.

What this path does

adl/src/production_birthday.rs is a thin wrapper that assembles a `ProductionBirthdayInput` and delegates to the kernel store. Its parameters are `Verified*` newtypes (`VerifiedMemoryPalaceAuthority`, `VerifiedResidentCycle`, `VerifiedToolAuthorityBinding`, `VerifiedBirthWitnessBinding`), so verification must have already happened for the function to be callable. The substantive logic lives in `adl-runtime-kernel/src/production_birthday.rs`.

Verified strengths

Crash-safe commit protocol, correctly sequenced

`write-intent` → `witness` → `staged receipt` → `atomic fs::rename` → `sync_directory` → `cleanup`. Every write uses `create_new(true)` so a stale staging file cannot be silently reused, and `sync_all` after each write. The directory `fsync` after the rename is present — the step most implementations omit, and without which the rename is not durable across power loss.

Idempotence and conflict detection are distinguished

`activate_owned` restores an existing receipt and returns it when `receipt_matches_input` holds; when input differs it returns `AlreadyCommitted` rather than overwriting. `recover_pending` separates “same transaction, resume” from “different transaction, conflict” via digest comparison of the pending intent.

Crash safety is tested by injection, not asserted

`ProductionBirthdayFailpoint` defines seven named interruption points (`BeforeIntent`, `AfterIntentSync`, `AfterWitnessSync`, `AfterReceiptSync`, `BeforeCommitRename`, `AfterCommitRename`, `AfterDirectorySync`). This is a proof surface capable of failing — the property repeatedly recommended in prior reviews and largely absent from earlier evidence artifacts.

Cross-binding validation actually compares

`validate_input` checks tool-authority schema, resident/cycle identity, continuity head, capability envelope, cognitive profile, and implementation revision for mutual agreement, and recomputes `candidate_digest` rather than trusting the supplied `packet_sha256`. This is the structural opposite of a status field asserting success for a check that compared nothing (cf. v0.91.7 `account_profile_validation_status`).

The non-claim is enforced in the type and validation layer

Activation is refused unless `birth_witness.packet().receipt.birth_event_status == BirthEventStatus::NotClaimed` and disposition is `WitnessesAccepted`. The system cannot activate while asserting a birth event occurred. The documented non-claim is encoded as a precondition rather than left to prose — the strongest form of claim discipline observed in this project.

Input hygiene

`validate_identifier` bounds length to 128 and restricts to `[A-Za-z0-9._-]`, preventing path traversal through resident-derived filenames. `is_sha256` enforces lowercase 64-hex.

`ProductionBirthdayStore::open` rejects a symlinked root. All serialization is JCS canonical (`serde_jcs`), so digests are stable across map ordering. All structs use `deny_unknown_fields`.

Findings

C1 · V092-CODE-01 — P2

title	Post-commit cleanup failure reports an error for a succeeded activation
evidence	<code>adl-runtime-kernel/src/production_birthday.rs</code> , <code>activate_owned</code> : after <code>fs::rename(&staging, committed_path)</code> succeeds, the subsequent <code>fs::remove_file(pending_path)</code> and <code>remove_file(witness_path)</code> each map failure to <code>ProductionBirthdayError::DurableWrite</code> and return.
impact	The birthday is durably committed, but the caller receives an error. A caller treating the error as failure could retry, alarm, or report non-activation for an activation that in fact occurred. Not corrupting: <code>recover_pending</code> detects the committed receipt on the next run, verifies input match, and clears the residue. Severity is limited to caller-visible state legibility.
fix	Introduce a distinct terminal variant (e.g. <code>CommittedWithCleanupPending</code>) returned when the rename succeeded but residue removal did not, so callers can distinguish committed-and-untidy from not-committed. Bounded: one enum variant plus two call-site branches.

C2 · V092-CODE-02 — P3

title	Integrity field defaults to empty string on serialization failure
evidence	<code>receipt_from_input</code> : <code>input_sha256</code> : <code>canonical_sha256(input).unwrap_or_default()</code> .
impact	On a serialization failure the receipt records an empty <code>input_sha256</code> . <code>receipt_matches_input</code> would then never match on a subsequent identical call, converting a resumable idempotent retry into a spurious <code>AlreadyCommitted</code> . Practically unreachable, since <code>validate_input</code> already performed the same serialization successfully on the same value; but <code>unwrap_or_default()</code> is the wrong default for an integrity field.
fix	Make <code>receipt_from_input</code> return <code>Result<ProductionBirthdayReceipt, ProductionBirthdayError></code> and propagate.

C3 · V092-CODE-03 — P3

title	Transaction lock is advisory and single-host; assumption is undocumented
evidence	<code>Ownership::acquire</code> uses <code>fs2::FileExt::try_lock_exclusive</code> on <code>{resident}.lock</code> .
impact	Correct for a single-host durable root. Advisory locking degrades on NFS and does not coordinate across containers or hosts sharing a mounted root, which would weaken the single-activation guarantee the store exists to provide. No current defect — a deployment constraint that is not stated in the source.
fix	Document the single-host local-filesystem assumption at the <code>ProductionBirthdayStore</code> type, and either reject known-network filesystems at <code>open()</code> or record the constraint in the deployment gate.

Cross-reference to documentation finding F2

`adl-runtime-kernel/src` measures 1.45 MB across 51 files. Within it, `continuity_control.rs` is 173 KB and `control.rs` is 155 KB — both larger than any file in the legacy `adl/src` crate except `long_lived_agent.rs` (219 KB) and `csm_runtime_api.rs` (211 KB). The module-size regrowth pattern flagged at v0.91.7 (when `csdlc-v2/store.rs` was 129 KB) has continued into the replacement kernel. The file reviewed here is 17 KB and readable; that is the standard worth generalizing. Recommend adopting an explicit module-size limit now, while the newer crates are still young.

Limitations

Read, not executed: no build, no tests, no failpoint runs, no audit, no git. Two files of a 51-file crate were read; no claim is made about the remaining 49, about `birth_witness.rs`, `birthday_identity.rs`, `memory_palace_authority.rs`, or about whether the `Verified*` constructors elsewhere in the crate enforce what their names imply — which is where the remaining trust in this path rests. Verifying those constructors is the natural next pass.